

Transformer-Based Intrusion Detection for IoT Networks

Uday Chandra Akuthota^{ID}, *Graduate Student Member, IEEE*, and Lava Bhargava^{ID}, *Senior Member, IEEE*

Abstract—Network intrusion detection systems are essential for defending recent computer networks from ever-evolving cyber attacks. Security is of utmost importance due to the complex and constantly changing nature of network threats. To improve the detection capabilities in network traffic, this research presents a unique method for intrusion detection by utilizing attention-based Transformer architectures. The proposed Transformer-based model offers an adaptable and reliable method for detecting sophisticated and dynamic threats by fusing the strength of the self-attention mechanism. The model is evaluated on two network intrusion benchmark datasets (NSL-KDD and UNSW-NB15). The correlation technique is used for feature extraction, and both binary and multiclass classification with and without feature extraction are performed on the datasets. The proposed model achieved over 99% accuracy, precision, and recall on the two datasets. The experimental results indicate that the proposed approach provides better results than other systems.

Index Terms—Feature extraction, intrusion detection, multihead attention (MHA), Transformer.

I. INTRODUCTION

NETWORK security issues are becoming more common in our lives due to the fast advancement of network technology, and their effects are becoming increasingly apparent. Presently, Internet resource utilization is increasing in daily life, but at the same time, there is a chance of security threats that are difficult to detect. After the introduction of the IoT, it is easy to process huge amounts of data rapidly through the network; such types of systems are more prone to attacks. The network intrusion detection approaches face more problems when finding network attacks. The primary challenge in finding attacks is improving the count and variety of attack types [1]. Traditional machine-learning methods in intrusion detection systems (IDSs) have not been demonstrated to be as adaptable and realistic since there is not enough traffic fluctuation or labeled data in the actual environment for some cases, such as variances in prediction accuracy. The frequency and intensity of network-based threats have increased during the past ten years. Developing novel techniques that leverage networks of communication, particularly wireless network systems, significantly contributes to the increase in network attacks. As a result, network security is crucial for preventing possible intrusions and assaults. This is achieved by employing security measures, defense systems, and network traffic analysis. A network IDS is usually installed on the network to

protect all connected devices by continually scanning network communication for unusual or malicious activity. Using this technique along with the anomaly driven IDS, an analysis of usual activity is created, and activities that differ from this characterization are identified as intrusions [2].

In network communication, the traffic data will appear sequentially. Existing machine learning (ML)-based IDSs frequently disregard sequential information to identify single network traffic data separately. This happens mainly because processing data sequentially with typical ML algorithms introduces specific challenges. Conversely, Transformer designs offer an efficient way to apply ML to detect attacks [3]. Furthermore, it is necessary to consider the network's long-term behavior and features to identify a significant percentage of intrusions in network activity. Even if intrusions are spread over a lengthy period, the Transformer's capacity to acquire long-range data can aid in detecting complicated network activity patterns that might suggest an impending attack. As a result, researching Transformer structures in the framework of network intrusion detection has a lot of potential. Using Transformers within the network environment is more complex than in the natural language processing domain. Transformers are well-established structures for managing text-based information, which is difficult in the network traffic domain. Several essential steps need to be taken to process networking input and classify using the Transformer's outcome, irrespective of the type of Transformer architecture. An overview of the study's primary contributions is provided below.

- 1) A Transformer-based novel model adept at accurately detecting attacks by efficiently parsing input characteristics into equivalent tokens for concurrent analysis.
- 2) The proposed model evaluation was performed on the two benchmark datasets with and without feature extraction, and the model's effectiveness was compared with the most recent research studies.

II. RELATED WORK

This section summarizes the comparison of signature and anomaly based detection, analysis of Transformer with traditional ML models, and several previously published research on network IDSs. IDS are essential cybersecurity technologies that keep an eye on the system activity and network traffic for unusual activity. Anomaly based IDS and signature-based IDS are two popular IDS techniques [4]. By evaluating incoming data to predetermined patterns or signatures of known attacks, signature-based IDSs are able to identify intrusions. It has a store of known attack signatures and employs pattern

Received 22 June 2024; revised 14 December 2024; accepted 31 December 2024. Date of publication 2 January 2025; date of current version 21 February 2025. (Corresponding author: Uday Chandra Akuthota.)

The authors are with the Department of Electronics and Communication Engineering, Malaviya National Institute of Technology, Jaipur 302017, India (e-mail: 2019rec9546@mnit.ac.in; lbhargava.ece@mnit.ac.in).

Digital Object Identifier 10.1109/JIOT.2025.3525494

TABLE I
COMPARISON OF SIGNATURE AND ANOMALY BASED DETECTION

Method	Metric	Merits	Best Application domain	Impediments
Signature based IDS	Detection accuracy	Reliable and accurate for known attacks.	Networks with predictable and well-documented threats.	Ineffective against unknown or modified attack patterns
	False Positives	Low, due to precise matching.	Critical systems where false alarms must be minimized.	Limited to detecting only predefined threats.
	False Negatives	Low for attacks that match existing signatures.	Static networks with consistent traffic behaviors.	High for unknown attacks.
	Scalability	Scales well for systems with fewer, well-defined threats.	Small or static networks with fewer changes in traffic.	Signature database size impacts efficiency in large networks.
	Resource usage	Lightweight and resource-efficient.	Resource constrained systems with predictable attack profiles.	Not adaptive to dynamic network changes.
	Maintenance	Simple with regular updates.	Environments with frequent, standardized updates.	Requires continuous updates to remain effective.
	Novel attack detection	NA	NA	Not applicable (NA) as it cannot detect unknown threats.
Anomaly based IDS	Detection accuracy	Effective for new or unseen threats.	Highly dynamic networks with evolving patterns	Accuracy depends on quality of training data.
	False Positives	Higher, as unusual legitimate behaviors might be flagged.	Networks where novel attacks need detection.	Frequent false positives reduce operational efficiency.
	False Negatives	Low for novel or unknown attacks.	Real-time systems requiring adaptive detection mechanisms.	Risk of overfitting if training is inadequate.
	Scalability	Scales effectively when supported by advanced ML models.	Large, dynamic networks with diverse behaviors.	Scalability depends on model complexity and computational power.
	Resource usage	Resource-intensive during training and operation.	High-resource environments or setups with dedicated monitoring.	High computational overhead can hinder real-time detection.
	Maintenance	Adaptive to evolving patterns through retraining.	Environments with rapidly changing normal behaviors.	Retraining is time-consuming and requires expert oversight.
	Novel attack detection	Effective in identifying novel and previously unseen attacks.	Networks prone to new or stealthy attack vectors.	Can produce false positives if the model lacks quality training data.

TABLE II
ANALYSIS OF TRANSFORMER AND TRADITIONAL ML MODELS

Aspect	Pros	Cons
Feature Representation	Rich contextual embeddings from Transformers Captures temporal and spatial dependencies effectively	Transformer training is resource-intensive. Poor feature extraction affects the entire pipeline.
Overfitting	Simpler designs reduce overfitting on smaller datasets in traditional ML models.	Dependence on Transformer output quality for effective generalization.
Interpretability	Traditional ML models allow explainability through feature importance methods	Transformer's internal representations are harder to interpret directly.
Computational Efficiency	Faster inference makes traditional ML models viable for edge deployment.	Transformer feature extraction is computationally expensive.
Loss of Information	Dimensionality reduction simplifies classification.	Risk of discarding essential features during dimensionality reduction.
Real-Time Suitability	Traditional ML models are lightweight and efficient during inference.	Training phase is slow due to Transformer's high resource demands.

matching to detect malicious activity. This approach excels at effectively recognizing known threats, with a low false positive rate, and is simple to use, as it requires frequent updates to remain efficient. However, it fails to identify novel or zero-day attacks, requires regular database updates, and has scalability issues as the signature database expands. It is widely used in antivirus software and firewalls, making it ideal for contexts with predictable and recognized threat vectors. In contrast, anomaly based IDS detects possible intrusions by identifying anomalies from established norms or standard behaviors. It simulates typical network activity during the training process and detects notable deviations as possible attacks. This adaptive method is particularly successful at identifying unknown attacks, including zero-day threats, and is ideal for dynamic or high-risk contexts, such as IoT networks and cloud systems. However, anomaly based detection has a high false positive rate since unexpected but valid activity might generate alerts. Furthermore, baseline modeling and configuration necessitate considerable computational resources and expertise.

The comparison of the two methodologies provides significant advantages and disadvantages. Signature-based IDS is best suited for stable environments with well-known threats since it provides high accuracy with few false positives. In contrast, anomaly based IDS is more suitable for situations with dynamic risks, enabling effective defense against new attacks but necessitating more complicated administration. Combining both techniques allows organizations to provide full protection. Table I describes the comparison of signature and anomaly based detection techniques.

A hybrid technique that combines Transformers and standard ML approaches is a potential option for conditions like as network intrusion detection [5]. In this configuration, the Transformer model analyses raw or semi-processed input data to extract high-level features, taking advantage of its capacity to capture complex temporal and spatial patterns. These features are then fed to a traditional machine-learning model for final categorization. This integration takes the use of Transformers' deep feature extraction capacity as well as

the simplicity and interpretability of traditional ML models. In the intrusion detection mechanism, the process generally consists of providing network traffic data into the Transformer, which includes flow-based information, such as timestamps, protocols, and port numbers. The Transformer encodes the sequential dependencies and generates high-quality feature vectors, which are then classified using a traditional machine-learning model. The hybrid approach has several advantages and disadvantages, which are provided in Table II.

The ensemble ML approach used in [1], provided better outcomes with the single classifier, and with the ensemble method the detection accuracy is reduced. A two-stage process proposed in [6], naive Bayes (NB) with an elliptical envelope (EE), where in the preliminary stage, data samples are categorized with ML models, and these classified samples are selected with majority voting technique. In the next stage, the EE approach classifies the detected attacks. A manifold regularized broad learning system with Lu decomposition (LU-MRBLS) was proposed in [7] to overcome the issues of the broad learning system. The potential information is extracted from the features using a manifold framework, and the computational error is reduced with the Lu-decomposition method, but this approach is suitable only for supervised learning techniques. The logistic regression with artificial bee colony (LR-ABC) approach used in [8] to reduce the training time of the detection model, the multiclassification is used for detecting each attack, and the training time was improved, but the detection accuracy is limited to 90.11% and 88.95% on NSL-KDD and UNSW-NB15 datasets. In [4], a genetic algorithm-based feature selection with the XGB (XgBoost) method was used to classify attacks. The model does not provide better outcomes for all the attack types. The contrastive learning approach proposed in [9], based on random masking, which can generate the relation between packet sequences. One-dimensional convolutional neural network (CNN1D) and deep neural network methods applied on the NSL and UNSW datasets provided an accuracy of 89.02% and 92.99%. In [10], an improved autoencoder (IAE) technique is utilized to detect the attacks, and binary classification is done on the datasets

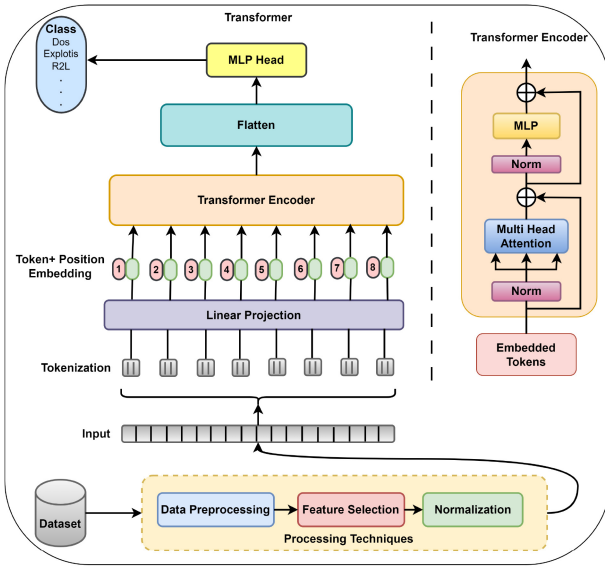


Fig. 1. Intrusion detection process using Transformer.

based on the provided results; improvement is required for the model. The deep learning-based feature selection (FS-DL) utilized in [11], for detecting attacks in SDN network, features are selected with association rule mining technique, but a very small number of feature sets are on both datasets for the classification. A hybrid CNN-Transformer method is suggested for intrusion detection. The Transformer is used to learn feature associations, whereas CNN is used to learn sample features [5]. The LSTM approach is used for the analysis of network data on both datasets, which produced an accuracy of 86.93% and 73.01% on NSL-KDD and UNSW-NB15, respectively, [12]. In [13], integrated ResNet, Transformer, and BiLSTM methods to detect intrusions by capturing both spatial and temporal network traffic features and leveraging their robust data representation learning capabilities, and in [14], used deformable ViT based on feature fusion to detect the intrusions. In [15], different encoding and classification techniques are applied to the Transformer model to increase network analysis.

This research introduces a Transformer-based approach that efficiently classifies the network information to detect the attacks, with essential elements like tokenization, Transformer encoder, and classification head, allowing for a Transformer-based network IDS.

III. METHODOLOGY

The proposed intrusion detection process using the Transformer model is shown in Fig. 1. Each block used in the proposed system is described below.

A. Processing Techniques

The datasets consist of raw data, and data processing is required before inputting it into the model. In data preprocessing data cleaning and one hot encoding technique are applied to transform the raw data into a suitable format for analysis. The correlation feature selection approach is applied

to the two datasets, and the Pearson's correlation coefficient matrix is computed for the features. The features with values greater than 0.5 are extracted from the total features. The normalization is performed with a min-max scaling technique to rescale the values. The processed input is fed to the proposed Transformer model.

B. Tokenization and Linear Projection

Since the Transformer takes input in the form of tokens, the input $x_i \in \mathbb{R}^{(1,d)}$ is divided into multiple small vectors to produce a token of dimensions $\mathbb{R}^{(1,d_1)}$, where $d_1 < d$, the computed tokens are projected into a latent space by multiplying with a linear operator E as $z_m = E x_m$, the main goal of these linear projections is to convert data into higher or lower-dimensional spaces so that the model may acquire more accurate token representations or associations from the input patterns. Position embeddings P are added to the token projected into the latent space to incorporate the position information as described in (1), and these are fed to the Transformer encoder

$$z_0 = [E_{x_1}, E_{x_2}, \dots, E_{x_m}] + P. \quad (1)$$

C. Transformer Encoder

The Transformer encoder is an essential part that enables the model to analyze sequences of inputs by exploiting self-attention processes, recording token relationships, and learning hierarchy representations. The tokens are routed via the proposed Transformer architecture comprising L encoder layers. Each layer l is made up of the following multihead attention (MHA) shown in (2), normalization (Norm), and multilayer Perceptron (MLP) blocks described in (3). Each MHA and MLP block has skip connections represented by addition in

$$y_m^l = \text{MHA}(\text{Norm}(z_m^l)) + z_m^l \quad (2)$$

$$z_m^{l+1} = \text{MLP}(\text{Norm}(y_m^l)) + y_m^l. \quad (3)$$

1) *Multihead Attention*: The scaled dot product attention mechanism is performed in the attention layer. It computes the relative importance of the data sequence by calculating key (K), query (Q), and value (V) vectors shown in Fig. 2. The weight matrices W_0^Q , W_0^K , and W_0^V are initialized randomly with the input, and these matrices are adjusted by backpropagation approach. The input vectors are multiplied with W_0^Q , W_0^K , and W_0^V to generate Q , K , and V . The vectors Q and K are multiplied to get the attention weights corresponding to all the values in the sequence. The attention is computed by applying attention weights to V as expressed in (4), where d_Q is the dimensionality of the Q and K vectors

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q * K^T}{\sqrt{d_Q}}\right) * V. \quad (4)$$

The multihead self-attention (MH) is computed by concatenating (Conc) the attention from multiple attention heads, as shown in (5). The self-attention at each node is computed by multiplying the weighted matrices with Q , K , and V as

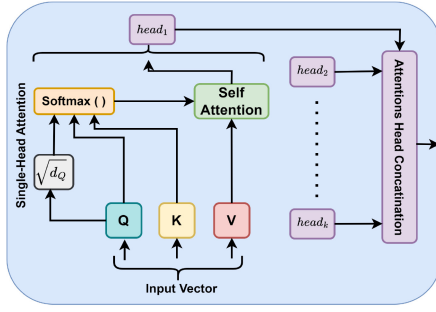


Fig. 2. Process flow map of MHA layer.

shown in (6), which is fed to the feed-forward network. Using this technique, the algorithm broadens its attention to multiple areas by learning important details in various representational subspaces

$$\text{MH}(Q, K, V) = \text{Conc}(\text{head}_1, \text{head}_2, \dots, \text{head}_k) \quad (5)$$

$$\text{head}_j = \text{Attention}\left(W_j^Q * Q_j, W_j^K * K_j, W_j^V * V_j\right). \quad (6)$$

2) *Normalization and MLP*: To stabilize the activations inside each layer of the Transformer, layer normalization is applied. Layer normalization is used independently after each sublayer output element. The output of MHA is fed to MLP, consisting of multiple linear layers separated by the ReLU activation function. The operation of MLP can be represented as shown in (7), where W_1 , W_2 and b_1, b_2 are learnable parameters

$$\text{FF}(x_i) = \text{ReLU}(x_i * W_1 + m_1) * W_2 + m_2. \quad (7)$$

D. Intrusion Classification

The intrusion classification is the final stage of the Transformer architecture. The MLP Head is used to classify intrusions, and the encoder output is given to the flattened layer to change the dimension of the input to 1d. The MLP Head comprises a pair of linear projections separated by a GELU nonlinearity, and the token dimensions, d , remain constant across every layer. The encoded input from the MLP head is classified with the softmax classifier into one of the output classes.

IV. EXPERIMENTS

This section uses accuracy, precision, and recall performance metrics to evaluate the proposed Transformer method on NSL-KDD and UNSW-NB15 datasets.

A. Dataset

The two network intrusion benchmark datasets are taken for the analysis, the UNSW-NB15 dataset consists of 257673 samples with 42 features and consists of DoS, exploits, reconssiance, analysis, normal, worms, backdoors, exploits, and fuzzer attacks. Where NSL-KDD dataset consists of 250000 samples with 44 features and consists of normal, Dos, R2L, probe, and U2R attacks.

B. Experimental Settings

The hyperparameter values for model architecture and training are detailed here. The proposed Transformer-based architecture comprises eight encoder layers with four heads in each MHA. The Adam optimizer is used to train the model, with a batch size of 16, embedded dimension 7, and a learning rate of 0.0001. The datasets used in the experiment consist of some classes with fewer samples. This class imbalance issue is overcome by using the focal loss function. The misclassification rate with focal loss is 0.74%, significantly lower than the 2.69% achieved with SMOTE, demonstrating better overall classification accuracy. Focal loss effectively handles class imbalance by emphasizing hard-to-classify samples during training, achieving superior performance without introducing synthetic data, unlike SMOTE, which may increase noise. The Transformer model is trained for 50 epochs.

C. Results

The experiments are performed for binary and multiclass classification with and without feature selection. The feature selection method discussed in Section III resulted in 25 and 15 features from NSL-KDD and UNSW-Nb15 datasets, respectively. The confusion matrices are shown in Fig. 3, indicates that the proposed model is able to detect intrusions reliably with high accuracy scores for binary and multiclass classification. The ROC curves for both datasets are shown in Fig. 4, and the proposed model evaluation outcomes with the comparison of other techniques are described in Table III. The model provided better outcomes with the advantage of the Transformer's self-attention mechanism and scalability. These allow the model to process all the data concurrently in a sequence and handle diverse and extensive data. The model provided the best results of 99.52%, 99.75%, 99.32%, and 99.53% in binary classification in terms of accuracy, precision, recall, and F1-score for NSL-KDD. For the UNSW-NB15 dataset, the model provides 99.97%, 99.98%, 99.95%, and 99.96% of accuracy, precision, recall, and F1-score for multiclass classification. The model provided better improvement in all performance metrics when compared with the other ML [1], [4], [6], [7], [8] and deep learning techniques [9], [10], [11]. Compared to other techniques, the model provides an accuracy improvement of around 6.12% on binary and 1.88% multiclass classification for NSL-KDD. For UNSW-NB15, the improvement of accuracy is 5.25% on binary and 7.83% on multiclassification compared to other ML and deep learning methods. Compared to [12], Transformers handle long-term dependencies more efficiently than LSTMs, using self-attention to capture global relationships across sequences. Unlike LSTM, which processes sequentially and struggles with very long inputs, Transformers handle them effectively without issues like vanishing gradient. The Transformer model demonstrated superior performance outcomes compared to similar approaches described in [5], [13], [14], and [15].

The total trainable parameters for the model is 13718. The computational power of the Transformer model involves

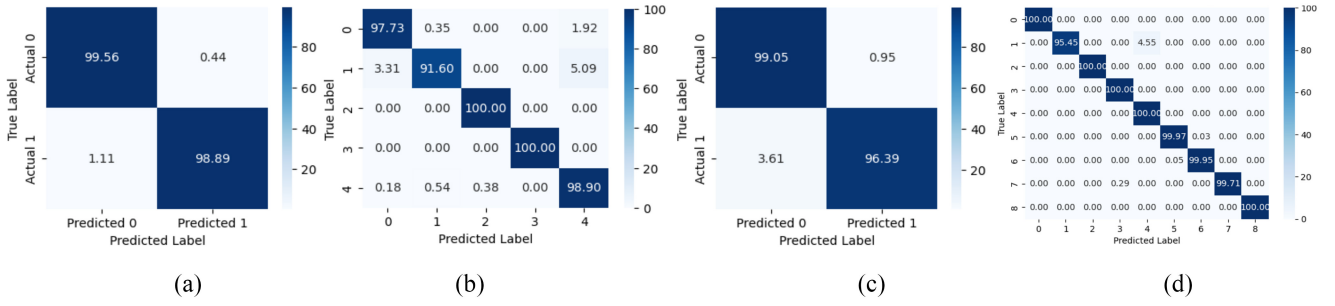


Fig. 3. Confusion matrices of binary and multiclass classification of NSL-KDD (a and b) and UNSW-NB15 (c and d) datasets, where classes 0 and 1 of binary classes indicate the normal and attack classes. In multiclass (0 = DoS, 1 = normal, 2 = probe, 3 = R2L, and 4 = U2R) for NSL-KDD and (0 = analysis, 1 = backdoor, 2 = DoS, 3 = exploits, 4 = fuzzers, 5 = generic, 6 = normal, 7 = reconnaissance, and 8 = worms) for UNSW-NB15.

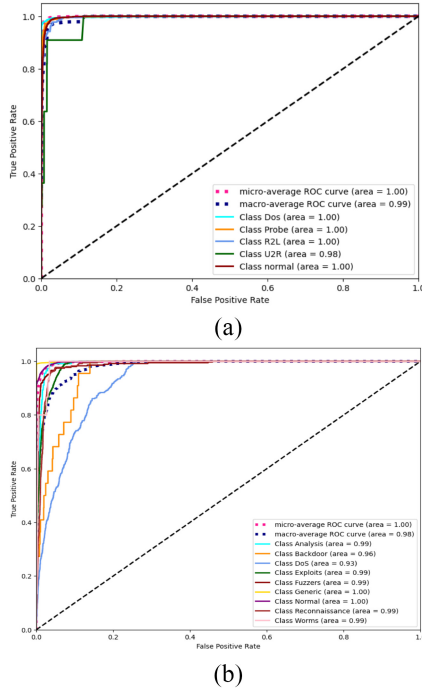


Fig. 4. ROC curves for the NSL-KDD (a) and UNSW-NB15 (b).

determining the number of floating point operations (FLOPs) required for a forward pass. The total FLOPs for the entire model is 84 176, and the time complexity of the Transformer model is, $O(L.(n.d^2 + n^2.d))$, where $n \rightarrow$ sequence length, $d \rightarrow$ Embedding dimension, and $L \rightarrow$ number of layers. Transformers handle large data volumes effectively due to their parallel processing, scalability, and high model capacity. They process sequences in parallel, enabling efficient computation and scalability to large datasets.

V. CONCLUSION AND FUTURE WORK

This work proposed a Transformer-based model that enables rapid development and assessment of Transformers for network IDSs. The model is evaluated with two real-time network intrusion benchmark datasets and compared with the other techniques used for the network IDSs. The proposed model outperforms the other models used for the intrusion detection. In the future, we intend to implement an intrusion

TABLE III
COMPARISON WITH THE RECENT LITERATURE

NSL-KDD							
Model	Year	F	Class	Acc(%)	Pre(%)	Rec(%)	F1s(%)
Ens_NB [1]	2021	20	Binary	88.10	95.90	82.60	88.70
CNN1D [9]	2022	-	Binary	89.02	89.11	89.70	88.98
XGB [4]	2023	16	Multi	95.54	92.61	95.54	93.41
FS-DL [11]	2023	4	Binary	91.47	-	-	-
LU-MRBLs [7]	2023	44	Binary	81.21	96.59	69.45	80.81
NB+EE [6]	2023	-	Multi	97.00	-	-	-
IAE [10]	2023	41	Binary	93.40	92.87	92.98	92.92
LSTM [12]	2023	22	Multi	86.93	-	-	-
Transformer [5]	2023	-	Multi	91.54	-	-	-
Res-TranBiLSTM [13]	2023	41	Multi	90.99	91.39	90.94	90.89
LR-ABC [8]	2024	-	Multi	90.11	-	-	-
Proposed	with FS	25	Binary	99.26	99.59	98.89	99.23
			Multi	98.88	99.64	98.96	99.29
	with out FS	44	Binary	99.52	99.75	99.32	99.53
			Multi	97.84	97.95	97.72	97.83
UNSW-NB15							
Ens_NN [1]	2021	19	Binary	85.70	80.80	93.50	86.70
DNN [9]	2022	-	Binary	92.99	92.93	92.90	92.91
XGB [4]	2023	24	Multi	89.58	89.46	89.58	88.89
FS-DL [11]	2023	6	Binary	87.48	-	-	-
LU-MRBLs [7]	2023	42	Binary	72.48	68.33	93.22	78.86
NB+EE [6]	2023	-	Multi	86.90	-	-	-
IAE [10]	2023	42	Binary	93.26	91.40	99.35	95.21
LSTM [12]	2023	17	Multi	73.01	-	-	-
LR-ABC [8]	2024	-	Multi	88.25	-	-	-
Flow Transformer [15]	2024	-	Binary	-	-	-	90.45
DE-ViT [14]	2024	-	Binary	97.25	97.1	-	-
Proposed	with FS	15	Binary	98.23	98.42	94.16	96.24
			Multi	97.41	99.41	95.21	97.26
	with out FS	42	Binary	98.41	96.98	96.39	96.68
			Multi	99.97	99.98	99.95	99.96

detection model on a platform like edge devices with smaller resource requirements.

REFERENCES

- [1] S. Das et al., "Network intrusion detection and comparative analysis using ensemble machine learning and feature selection," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 4, pp. 4821–4833, Dec. 2022.
- [2] L. Qi, Y. Yang, X. Zhou, W. Rafique, and J. Ma, "Fast anomaly identification based on multispect data streams for intelligent intrusion detection toward secure industry 4.0," *IEEE Trans. Ind. Informat.*, vol. 18, no. 9, pp. 6503–6511, Sep. 2022.
- [3] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Sep. 2022.
- [4] A. Srivastava, D. Sinha, and V. Kumar, "WGAN-GP based synthetic attack data generation with GA based feature selection for IDS," *Comput. Secur.*, vol. 134, Nov. 2023, Art. no. 103432.
- [5] R. Yao, N. Wang, P. Chen, D. Ma, and X. Sheng, "A CNN-transformer hybrid approach for an intrusion detection system in advanced metering infrastructure," *Multimedia Tools Appl.*, vol. 82, no. 13, pp. 19463–19486, 2023.
- [6] M. Vishwakarma and N. Kesswani, "A new two-phase intrusion detection system with naïve bayes machine learning for data classification and elliptic envelop method for anomaly detection," *Decision Anal. J.*, vol. 7, Jun. 2023, Art. no. 100233.
- [7] Y. Liu, K. Zhang, and Z. Wang, "Intrusion detection of manifold regularized broad learning system based on LU decomposition," *J. Supercomput.*, vol. 79, pp. 1–49, Dec. 2023.

- [8] B. Kolukisa, B. K. Dedetürk, H. Hacilar, and V. C. Gungor, "An efficient network intrusion detection approach based on logistic regression model and parallel artificial bee colony algorithm," *Comput. Stand. Interfaces*, vol. 89, Apr. 2024, Art. no. 103808.
- [9] Y. Yue, X. Chen, Z. Han, X. Zeng, and Y. Zhu, "Contrastive learning enhanced intrusion detection," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 4, pp. 4232–4247, Dec. 2022.
- [10] H. Yu, C. Kang, Y. Xiao, and Y. Ting, "Network intrusion detection method based on hybrid improved residual network blocks and bidirectional gated recurrent units," *IEEE Access*, vol. 11, pp. 68961–68971, 2023.
- [11] L. Zhang, K. Liu, X. Xie, W. Bai, B. Wu, and P. Dong, "A data-driven network intrusion detection system using feature selection and deep learning," *J. Inf. Secur. Appl.*, vol. 78, Nov. 2023, Art. no. 103606.
- [12] S. M. Kasongo, "A deep learning technique for intrusion detection system using a recurrent neural networks based framework," *Comput. Commun.*, vol. 199, pp. 113–125, Feb. 2023.
- [13] S. Wang, W. Xu, and Y. Liu, "Res-TranBiLSTM: An intelligent approach for intrusion detection in the Internet of Things," *Comput. Netw.*, vol. 235, Nov. 2023, Art. no. 109982.
- [14] K. He, W. Zhang, X. Zong, and L. Lian, "Network intrusion detection based on feature image and deformable vision transformer classification," *IEEE Access*, vol. 12, pp. 44335–44350, 2024.
- [15] L. D. Manocchio, S. Layeghy, W. W. Lo, G. K. Kulatilleke, M. Sarhan, and M. Portmann, "FlowTransformer: A transformer framework for flow-based network intrusion detection systems," *Expert Syst. Appl.*, vol. 241, May 2024, Art. no. 122564.